Computer Programming 143 – Lecture 11
Functions III

Faculty of Engineering

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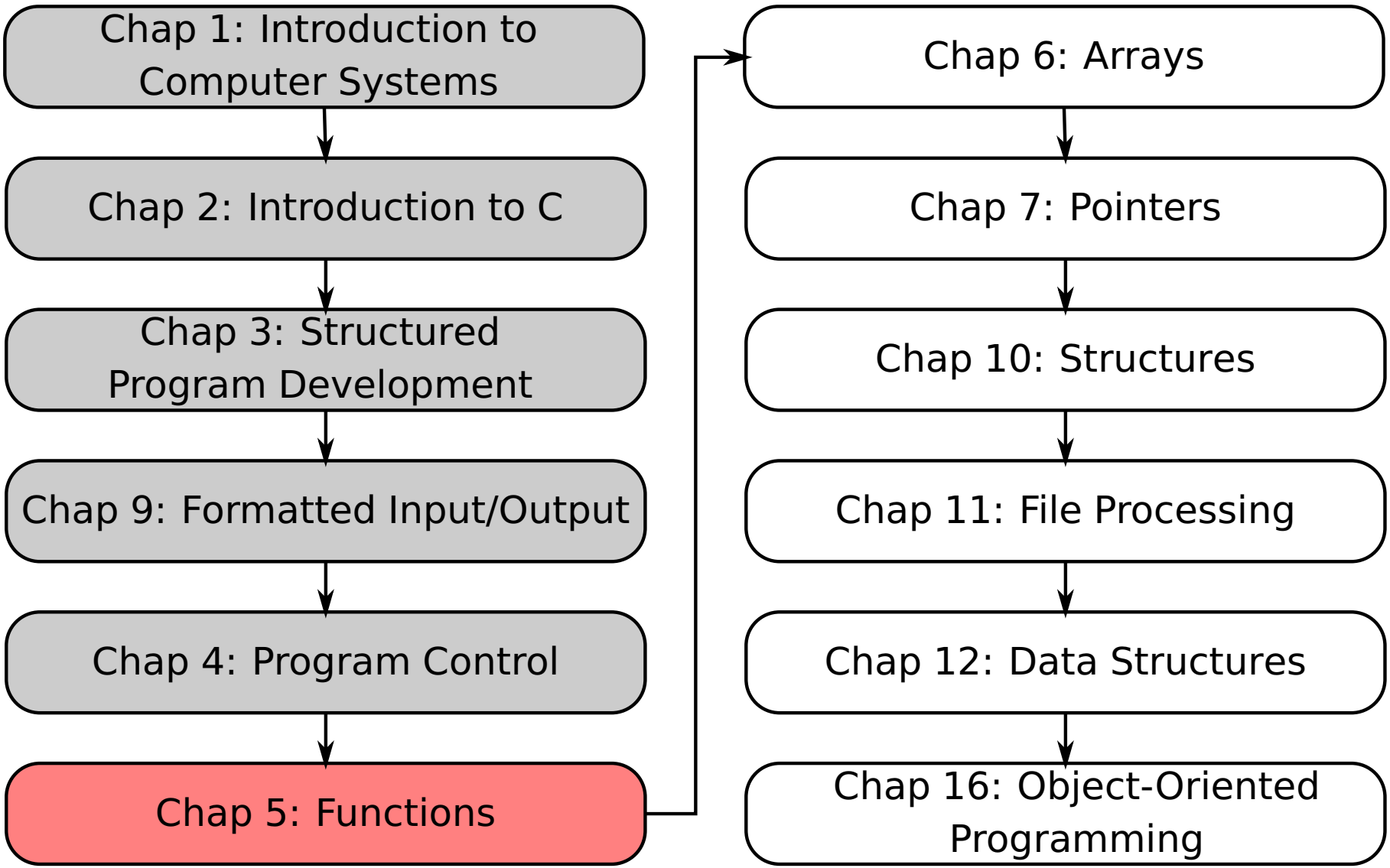
Module Overview



Lecture Overview

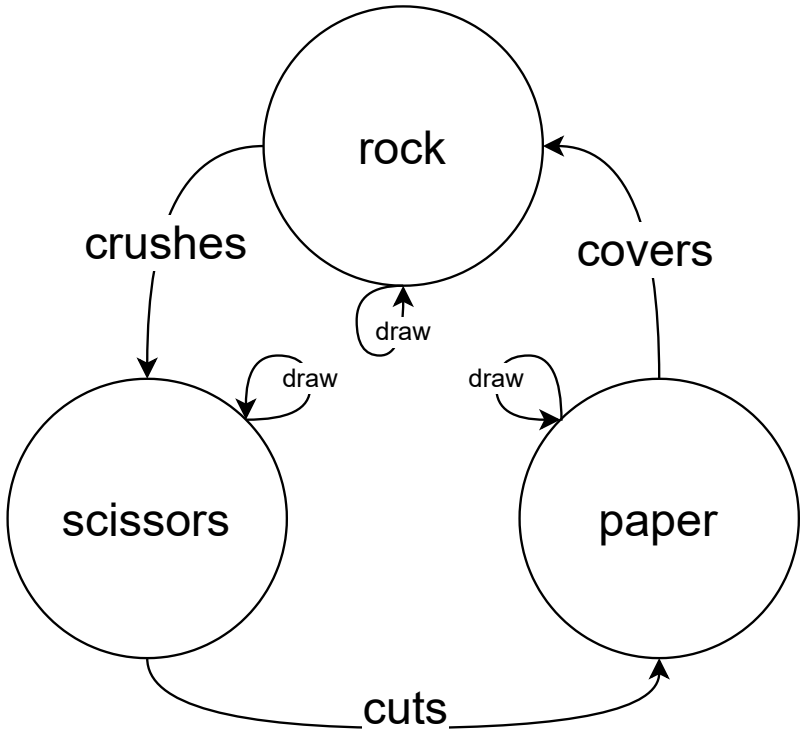
1. Example: A Game of Chance (5.11)
2. Storage Classes (5.12)
3. Scope Rules (5.13)

5.11 Example: A Game of Chance I

Rules of rock, paper, scissors

Each player simultaneously forms one of three shapes with an outstretched hand. The shapes are "rock", "paper", and "scissors". A simultaneous, zero-sum game, it has three possible outcomes: a draw, a win, or a loss. "rock" breaks (wins) "scissors", "scissors" cuts (wins) "paper", "paper" covers (wins) "rock". If both players choose the same shape, the game is a tie.

5.11 Example: A Game of Chance II



C code

Refer to Fig. 5.7 in Deitel & Deitel

5.11 Example: A Game of Chance III

C code

```c
// fig05_07.c
// Simulating the game of rock, paper, scissors
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <time.h> // contains prototype for function time

enum GameStatus {CONTINUE, GAME_WON, GAME_LOST}; // constants represent g
// status
enum RoundStatus {DRAW, WON, LOST}; // constants represent round status
enum Shape {ROCK, PAPER, SCISSORS}; // constants representing possible sh
enum Shape computerPlayRandomShape(void); // playRandomShape function
// prototype;
enum RoundStatus getRoundStatus(enum Shape, enum Shape);
enum Shape convertIntToShape(int);
char* getStringFromShape(enum Shape);
```

5.12 Storage Classes I

Storage class specifiers

An identifier's storage class determines:

- **Storage duration** – how long a variable exists in memory
- **Scope** – where object can be referenced in program
- **Linkage** – specifies the files in which an identifier is known (more in Chapter 14)

5.12 Storage Classes II

Local variables

- Variables exist for certain section of program (block of code)
- Exist while block is active
- Destroyed when block is exited

Static storage

- Variables exist for entire program execution
- **`static`**: local variables defined in functions.
  - Keep value after function ends
  - Only known in their own function
- Global variables and functions
  - Known in any function

5.12 Storage Classes III

Example

```
void accumulate(void)
{
     static int acc = 0;  // static variable definition
     int num;  // local variable definition
     printf("Enter a number: ");
     scanf("%d", &num);
     acc += num;  // add number to current total
     printf("The current total is %d\n", acc);
}
```

5.13 Scope Rules I

Definition of scope

Portion of the program in which the identifier can be referenced

File scope

- Variable defined outside a function
- Known in all functions (in file)
- Used for global variables, function definitions, function prototypes

Block scope

- Variable declared inside a block (`{...}`)
- Used for variables, function parameters (local variables of function)
- Outer blocks “hidden” from inner blocks if there is a variable with the same name in the inner block

5.13 Scope Rules II

Example

```c
/* Example indicating scope of variable */
#include <stdio.h>

void useLocal(void);  // function prototype
void useStaticLocal(void);  // function prototype
void useGlobal(void);  // function prototype

int x = 1;  // global variable
```

5.13 Scope Rules III

Example (cont...)

```
int main(void)
{
    int x = 5;  // local variable to main

    printf("local x in scope of main is %d\n", x);
    useLocal();  // useLocal has local x
    useStaticLocal();  // useStaticLocal has static local
    useGlobal();  // useGlobal uses global x
    useLocal();  // useLocal has local x
    useStaticLocal();  // useStaticLocal has static local
    useGlobal();  // useGlobal uses global x
    printf("local x in scope of main is %d\n", x);

    return 0;
}
```

5.13 Scope Rules IV

Example (cont...)

```
// useLocal reinitializes local variable x during each call
void useLocal(void)
{
    int x = 25;  // initialized each time useLocal is called
    printf("\nlocal x in useLocal is %d ", x);
    printf("after entering useLocal\n");
    ++x;
    printf("local x in useLocal is %d ", x);
    printf("before exiting useLocal\n");
}
```

5.13 Scope Rules V

Example (cont...)

```
/*
useStaticLocal initializes static local variable x only
the first time the function is called;
value of x is saved between calls to this function
*/
void useStaticLocal(void)
{
    static int x = 50;  // initialized only once
    printf("\nlocal x in useStaticLocal is %d ", x);
    printf("after entering useStaticLocal\n");
    ++x;
    printf("local x in useStaticLocal is %d ", x);
    printf("before exiting useStaticLocal\n");
}
```

5.13 Scope Rules VI

Example (cont...)

```
// function useGlobal modifies global variable x during each call
void useGlobal(void)
{
    printf("\nglobal x is %d after entering useGlobal\n", x);
    x *= 10;
    printf("global x is %d before exiting useGlobal\n", x);
}
```

Use globals sparingly! Preferably never

5.13 Scope Rules VII

Output:

```
local x in scope of main is 5

local x in useLocal is 25 after entering useLocal
local x in useLocal is 26 before exiting useLocal

local x in useStaticLocal is 50 after entering useStaticLocal
local x in useStaticLocal is 51 before exiting useStaticLocal

global x is 1 after entering useGlobal
global x is 10 before exiting useGlobal
```

5.13 Scope Rules VIII

Output: (cont...)

```
local x in useLocal is 25 after entering useLocal
local x in useLocal is 26 before exiting useLocal

local x in useStaticLocal is 51 after entering useStaticLocal
local x in useStaticLocal is 52 before exiting useStaticLocal

global x is 10 after entering useGlobal
global x is 100 before exiting useGlobal

local x in scope of main is 5
```

Perspective

Today

Functions III

- Example: a game of chance
- Storage class
- Scope rules

Next lecture

Functions IV

- Recursion

Homework

1. Study Sections 5.11-5.13 in Deitel & Deitel
2. Do Self Review Exercise 5.6 in Deitel & Deitel
3. Do Exercise 5.27 in Deitel & Deitel